

FLAGSHIP SOLUTION SCALING TRACK PROFILE

Scaling Inclusive Mechanization through Rural Entrepreneurship and Financial Innovation

Scaling for Impact Science Program | Area of Work 2

Grand Challenge 2 · Deliver Inputs and Services at Scale through Inclusive Market Systems

Flagship 2.2 · Entrepreneurship through Mechanization & Postharvest Services

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Foreword

CGIAR SCALING FOR IMPACT · AREA OF WORK 2 — PATHWAYS TO SCALE IN AGRIFOOD SYSTEMS

CGIAR produces far more scale-ready innovation than currently reaches farmers at scale. Closing that gap is the purpose of the CGIAR Scaling for Impact (S4I) Science Program. Within S4I, Area of Work 2 — Pathways to Scale in Agrifood Systems — is organized around three Grand Challenges that mark where scaling can unlock the largest development returns across food, land, and water systems in a climate crisis. These Grand Challenges were defined through an evidence-led process — drawing on the CGIAR Impact Report 2022–2024, Initiative reporting, and an analysis of eleven major donor strategies — so that each rests on demonstrated scaling potential and connects directly to the priorities of CGIAR's funders.

Under each Grand Challenge sits a focused set of **Scaling Flagships** that "dock with" and support CGIAR's Science Programs — taking mature innovation bundles and accelerating their delivery and scaling systems:

- **Grand Challenge 1 — Close the Climate Adaptation Gap** through scalable climate services and smart water management

Flagship 1.1 — Digital Climate Services for Resilient Farm & Enterprise Management

Flagship 1.2 — Scaling Smart Irrigation & Climate-Resilient Water Systems

- **Grand Challenge 2 — Deliver Inputs and Services at Scale** through inclusive market systems

Flagship 2.1 — Scaling Resilient, Nutritious & Biodiverse Seed & Soil Systems

Flagship 2.2 — Entrepreneurship through Mechanization & Postharvest Services

Flagship 2.3 — Genetics, Inputs & Services for Animals & Aquaculture

- **Grand Challenge 3 — Drive Demand, Shape Behavior & Activate Policy** for nutrition

Flagship 3.1 — Catalyzing Demand for Nutritious Diets through Markets, Institutions & Policy

Each Flagship is delivered through **Scaling Solution Tracks** — the profile that follows is one of them. Tracks are run deliberately as large, time-bound scaling campaigns, not as research projects or pilots. Experience across the portfolio shows that the binding constraint to scaling is rarely the innovation itself; it is *system integration* — the delivery and business models, finance and risk-sharing, institutional and policy alignment, and coordination among actors required to move a proven solution into widespread use. Each track therefore bundles a validated innovation with a clear delivery pathway, committed partners, and a defined geography, and is driven through and with scaling partners rather than by CGIAR alone — leveraging complementary investments to amplify reach.

This campaign approach lets S4I concentrate resources, operate in performance-based mode against measurable KPIs — farmers adopting innovations, jobs created, consumers gaining access to healthier diets, and scaling finance and supportive policy mobilized — and hold itself accountable to the S4I 2030 outcome targets and CGIAR's Impact Area goals. Across every track, scaling is pursued responsibly and inclusively, with women, youth, and marginalized groups systematically included by design.

1. Overview

Agriculture in Southern Africa remains heavily reliant on manual labour and animal draft power, a model increasingly disrupted by climate change, disease, and labour shortages. In Zambia, recurring droughts have decimated draft cattle herds, leaving smallholder farmers, particularly women and youth, without reliable traction for timely land preparation, planting, and postharvest processing. This limits the adoption of climate-smart practices, suppresses yields, and perpetuates rural drudgery. This Scaling Track addresses this critical bottleneck by establishing a sustainable and inclusive mechanization ecosystem.

The primary goal is to scale access to climate-smart, scale-appropriate mechanization for smallholder farmers by catalyzing a network of rural entrepreneurs. The solution is a bundled "Mechanization Starter Pack" combined with a **Mechanization Service Provider (MSP)** delivery model. The bundle integrates: (i) access to low-cost, multi-purpose machinery (e.g., two-wheel tractors, rippers, direct seeders, shellers, and postharvest equipment); (ii) tailored financial products with risk mitigation; and (iii) comprehensive technical and business training. This approach transforms rural youth and women into paid service providers, creating enterprises while delivering affordable, pay-per-use services to their communities.

The scaling pathway is anchored in Zambia, with a strategic vision to expand to Zimbabwe, Malawi, Kenya, and Ethiopia. The model is not theoretical; it is built on a robust evidence base from Zimbabwe, where over 500 MSPs were established between 2022 and 2025, serving more than 20,000 farmers and generating over US\$40,000 in revenue, even during a drought year. This track is led by CIMMYT in close partnership with the Alliance of Bioversity International and CIAT (ABC) and the International Water Management Institute (IWMI), and is driven by strong demand from government ministries, financial institutions, and private-sector suppliers.

2. Innovation Bundle and Scaling Ambition

2.1 What is Being Scaled

- **Mechanization Service Provider (MSP) Model:** A private-sector-led approach where rural entrepreneurs acquire machinery and sell services to farmers.
- **Bundled "Starter Pack":** A curated set of scale-appropriate machinery, including two-wheel tractors, basin diggers, rippers, direct seeders, shellers, threshers, and trailers.
- **Innovative Financial Instruments:** Bundled loan products with climate risk insurance, supported by risk-sharing facilities (RSFs) to de-risk lending for financial institutions.
- **Service Ecosystem Support:** A package of technical training, financial literacy, business development support, and regional service hubs for spare parts and repairs.
- **Gender, youth and inclusion pathways:** Targeted support to enable women and youth participation as MSPs and service users, including tailored financing, capacity development, and labour-saving solutions.
- **Demand Creation and Market Linkages:** Structured farmer demand generation through demonstration-to-adoption pathways, combined with market linkages to aggregators, processors, and off-takers to ensure services translate into income.

2.2 Evidence of Impact

- **Proven Reach and Revenue:** In Zimbabwe (2022-2025), the model established **500+ MSPs** serving over **20,000 farmers** (with spillover to 35,000+), generating over **US\$40,000 in revenue** for MSPs despite a severe drought.

- **Private Sector Leverage:** The pilot raised over **US\$337,000** in private sector investment from service providers and financiers.
- **Strong Demand:** High willingness-to-pay confirmed through recent studies and Scaling Readiness workshops. Over 1,000 units of a popular no-till planter were sold in Zimbabwe in a single year after ecosystem elements were established.
- **Scalability Confirmed:** The innovation is assessed at Scaling Readiness Levels 5-6, having moved beyond proof-of-concept to proven operational success in multiple environments.

2.3 Scaling Targets

Mechanization Providers established	Service (MSPs)	500 (in Zambia)	1,000 (in Zambia); expanded to 4 additional countries
Smallholder farmers accessing mechanization services		10,000	>100,000 (regionally)
Investment (private and institutional)	leveraged	US\$ 500,000+	US\$ 2-3 Million+
Financial products developed/deployed		At least 1 new bundled product is operational in Zambia	Products mainstreamed by multiple financial institutions regionally
Hectares through MSPs	served	9,000	100,000 (regionally)
Decent rural jobs created (MSPs + operators + support services)		500	2,000
Inclusion metric (women and youth participation)		≥40	≥50

3. Scaling Pathway and Theory of Change

3.1 Pathway Strategy The core scaling strategy is a **blended public-private partnership (PPP) model** designed for long-term sustainability and market-based growth. It is not a subsidy model but a market activation strategy. The pathway is anchored in structured demand activation to ensure sustained utilization of services.

The pathway begins by de-risking the entry point for both entrepreneurs and financiers. **MSPs** are recruited with a focus on youth and women and contribute ~30% equity. **Financial institutions** (banks, SACCOs, MFIs) provide the remaining ~70% as a loan, bundled with climate risk insurance. A catalytic **Risk-Sharing Facility (RSF)** covers 25-65% of potential loan defaults, incentivizing banks to lend to this new market. Loan repayment schedules are aligned with agricultural crop cycles to minimize default risk.

To ensure operational sustainability, the pathway includes establishing **regional service hubs** that provide training, spare parts, and repair services, supported by trained rural mechanics. This embeds the service

ecosystem within local market systems. Finally, the pathway actively works to **embed the MSP model into government programs** (e.g., Zambia's National Mechanization Strategy) and private sector supply chains, ensuring institutional uptake and replication beyond the life of the project.

3.2 Theory of Change If smallholder farmers gain reliable access to affordable, climate-smart mechanization services through a network of viable rural enterprises (MSPs)... **and if** these enterprises are supported by fit-for-purpose financial products (loans, insurance, risk-sharing) and a robust ecosystem for skills training and equipment maintenance... **then** farmers will adopt more timely and efficient land preparation, crop establishment, and postharvest practices, leading to increased productivity, reduced drudgery, and improved resilience... **contributing to** higher household incomes, the growth of inclusive rural economies, and the large-scale transformation of food systems in Southern Africa.

3.3 Key Activities Funded by S4I

- **Ecosystem Activation:** Finalizing innovation bundles, co-designing financial instruments with banks, and operationalizing the Risk-Sharing Facility (RSF) with partners.
- **Entrepreneurship Development:** Recruiting, training, and equipping 500 MSPs in Zambia, with targeted inclusion of women and youth using tools like GenderUp.
- **System Building:** Establishing operational service hubs and training rural mechanics to ensure equipment uptime and create local employment.
- **Learning and Policy Engagement:** Conducting scaling readiness assessments, documenting the scaling pathway, and facilitating policy dialogues to embed the model into national programs.
- **Demand activation and Service Uptake:** Financing structured demand creation through demonstration platforms, lead farmer networks, and extension systems to stimulate farmer uptake and ensure consistent utilization of MSP services.

4. Partners and Roles

Demand and Delivery Partners			NGOs: Total LandCare; Conservation Farming Unit (CFU); Private Sector: CAMCO; SARO; Growmore; Aggregators: Multiple contract farming schemes.	NGOs: Last-mile delivery, MSP recruitment, training, and farmer mobilization.
Demand and Delivery Partners			NGOs: Total LandCare; Conservation Farming Unit (CFU); Private Sector: CAMCO; SARO; Growmore; Aggregators: Multiple contract farming schemes.	Private Sector: Machinery supply, after-sales service, and exploring distributor-finance and digital scheduling models.
Demand and Delivery Partners			NGOs: Total LandCare; Conservation Farming Unit (CFU); Private Sector: CAMCO; SARO; Growmore; Aggregators: Multiple contract farming schemes.	Aggregators: Provide entry points for structured repayment and co-financing.
Scaling and Enabling Partners			Government: Zambia Ministry of Agriculture (and counterparts in spillover countries); Financial Institutions: ZANACO, AgLeaseCo, Vision Fund, Microloan Foundation; Development Programs: Ukama Ustawi, SIFAZ, AICCRA, Water and Soil Accelerator.	Government: Policy alignment, embedding MSPs in extension and subsidy programs, and convening PPPs.
Scaling and Enabling Partners			Government: Zambia Ministry of Agriculture (and counterparts in spillover countries); Financial Institutions: ZANACO, AgLeaseCo, Vision Fund, Microloan Foundation; Development Programs: Ukama Ustawi, SIFAZ, AICCRA, Water and Soil Accelerator.	Financial Institutions: Design and deploy bundled loans, insurance, and manage the Risk-Sharing Facility (RSF).
Scaling and Enabling Partners			Government: Zambia Ministry of Agriculture (and counterparts in spillover countries); Financial Institutions: ZANACO, AgLeaseCo, Vision Fund, Microloan Foundation; Development Programs: Ukama Ustawi, SIFAZ, AICCRA, Water and Soil Accelerator.	Programs: Provide complementary funding, technical expertise, and regional scaling platforms.

Funding Model: S4I catalytic funding is used to de-risk market entry, build capacity, and foster partnerships; it does not directly finance machinery, which is covered by a blend of MSP equity (30%) and private/commercial loans (70%).

5. Enabling Environment and Risks

5.1 Opportunities

- **Strong Government Alignment:** Zambia recently launched a National Mechanization Strategy, creating a high-level political will and a clear entry point for policy integration.

- **Private Sector Appetite:** Machinery distributors and financial institutions in Zambia are more advanced in their interest in mechanization financing than in neighbouring countries, with several already exploring pilot lending schemes.
- **Leverageable Programs:** A strong portfolio of bilateral projects (SIFAZ, AICCRA, UU+, Water and Soil Accelerator) in the region provides co-investment and delivery platforms, multiplying the impact of S4I funds.
- **Proven Regional Model:** The success of the MSP model in Zimbabwe provides a powerful proof-of-concept, operational playbook, and a network of peer-learners for Zambian entrepreneurs.
- **Regional Expansion Potential:** Similar agro-ecological and market conditions across Zambia, Zimbabwe, Malawi, and Mozambique create strong potential for rapid replication once the model is proven.
- **Emerging Financial Innovation Ecosystem:** Growing experience with bundled financial products (credit + insurance + advisory) in the region provides a foundation for scaling mechanization finance models.

5.2 Challenges and Mitigation

- **Financial Risk:** Banks may perceive lending to smallholder-linked enterprises as high risk.
- *Mitigation:* Co-design Risk-Sharing Facilities (RSF) and bundled insurance; pilot with progressive banks; use aggregators for structured repayment.
- **Operational Bottlenecks:** Lack of spare parts and repair services can render machinery idle, destroying business viability.
- *Mitigation:* Establish regional service hubs from the outset; train a cadre of rural mechanics; work closely with suppliers to ensure after-sales service.
- **Inclusion Failure:** Without deliberate action, women and youth may be excluded from service provision roles.
- *Mitigation:* Apply inclusive scaling tools like GenderUp and GALS; target women and youth in recruitment; promote gender-responsive equipment (e.g., self-starting tractors).
- **Timing and Funding Cycles:** Annual funding cycles conflict with agricultural seasons, risking delays and missed opportunities.
- *Mitigation:* Clearly escalate urgency internally; use Q1 2026 for foundational work (partner MoUs, training curricula) that is not season-dependent.

6. Learning and Adaptation

6.1 Monitoring Focus

- **Process Outcomes:** Speed of MSP recruitment, training completion, and financial disbursement. Functionality of service hubs.
- **Behavior Change:** Adoption of pay-per-use services by farmers; business management skills of MSPs; willingness of banks to lend to the sector.
- **System Outcomes:** Shifts in government policy or programs to include MSPs; replication of the model by the private sector without CGIAR facilitation; evidence of gender-inclusive business growth.

6.2 Adaptive Management

Learning is embedded in the track's DNA. The team will use:

- **Quarterly Check-ins:** Formal reviews with the S4I program team to assess progress against milestones and discuss strategic pivots.

- **Scaling Readiness Assessments:** Periodic workshops to re-evaluate the maturity of the innovation package and adjust the scaling strategy as needed.
- **Feedback Loops:** Data from service hubs (on repairs, parts) and financiers (on repayment) will be used to provide real-time feedback to equipment designers and training modules.
- **GenderUp Reviews:** Continuous application of participatory methods to ensure the scaling process remains inclusive and responsive to diverse user needs.

7. Next Steps and Timeline (2026)

Q1	Finalize and Align: Secure budget codes and initiate spending. Finalize innovation bundles and financial instruments with partners. Sign MOUs with key government and financial partners. Develop detailed training curricula.
Q2	Recruit and Train: Launch awareness campaigns and recruit first cohort of MSPs (focus on women/youth). Initiate technical and business management training. Operationalize partnerships with machinery suppliers.
Q3	Finance and Equip: Facilitate loan applications and disbursements with partner financial institutions. Begin distribution of starter packs to trained and financed MSPs. Establish first regional service hubs.
Q4	Operationalize and Learn: Support MSPs in delivering services (focus on postharvest/transport services). Verify 500 MSP target via training records and asset tags. Conduct first learning review and document early lessons for 2027 scale-up.

Communications and Knowledge: The track will prioritize documenting the "how" of scaling through impact stories, policy briefs, and video content, focusing on the behavioral and systems changes that enable scale, rather than just final impact numbers. This will be complemented by structured knowledge products (e.g., scaling playbooks, case studies), cross-country learning exchanges, and targeted dissemination to policymakers, financial institutions, and private sector actors.

8. Contact(s)

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